

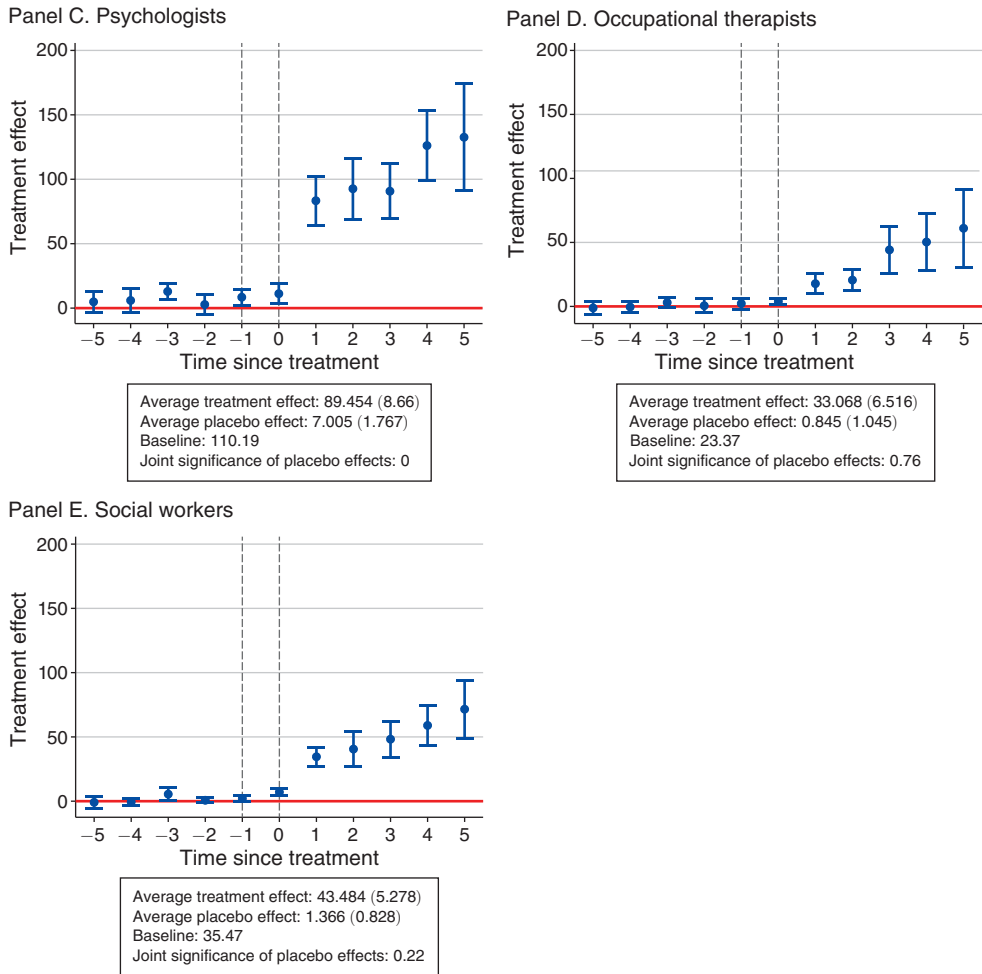


FIGURE 4. EFFECTS OF CAPS ON OUTPATIENT CARE BY MENTAL HEALTH SPECIALISTS: NUMBER OF OUTPATIENT PROCEDURES PERFORMED BY EACH TYPE OF SPECIALIST PER 10,000 PEOPLE (*continued*)

Notes: This figure plots 95 percent confidence intervals computed with a municipality-level clustered bootstrap and dynamic and placebo DID estimators for the CAPS' effects on the number of outpatient procedures per 10,000 people performed by mental health practitioners. We present the effects on procedures performed by any mental health professionals—the sum of the next categories (panel A), psychiatrists (panel B), psychologists (panel C), occupational therapists (panel D), and social workers (panel E). Placebo DID estimators estimate the CAPS' effects had the treatment occurred in a placebo, pretreatment period. They use a varying baseline period—the one immediately before the placebo treatment date—so we do not normalize relative to a unique period. Controls include municipality GDP per capita, PBF spending per capita, a series of indicators for age-by-gender population bins, state $\times$ year fixed effects, and linear trends of pretreatment characteristics. Pretreatment municipality characteristics included are: Theil index, poverty rate, unemployment rate, illiteracy rate, share of rural population, log of population, log of social spending, number of mental health providers, number of mental health offices, municipality area, altitude, distance to capital, temperature, and rainfall. The Average Treatment Effect computes a weighted average of the dynamic estimators, giving to each estimator a weight proportional to the number of switchers the estimator applies to. In parentheses, standard errors are computed with a municipality-level clustered bootstrap. Baseline indicates the sample mean values for the treated in the first pre-CAPS period.

effect is -0.948 (s.e., 0.28) or 7.5 percent relative to baseline. Our results further indicate that the policy reform is mainly associated with reductions in long-stay

TABLE 2—EFFECTS OF CAPS ON OUTPATIENT CARE BY MENTAL HEALTH SPECIALISTS AND FOR SELECTED PROCEDURES: NUMBER OF OUTPATIENT PROCEDURES PER 10,000 PEOPLE

	2-year effect (1)	5-year effect (2)	Placebo effect (3)	Mean at baseline (4)
<i>Outpatient services by MH practitioners</i>				
Total	206.754 (24.357)	264.213 (27.050)	11.548 (4.05)	233.683
Psychiatrists	102.864 (12.801)	98.209 (10.267)	2.331 (1.275)	64.644
Psychologists	62.409 (8.647)	89.454 (8.660)	7.005 (1.767)	110.192
Therapists	14.089 (4.889)	33.068 (6.516)	0.845 (1.045)	23.375
Social workers	27.392 (5.283)	43.484 (5.278)	1.366 (0.828)	35.473
<i>Specific outpatient services</i>				
Antipsychotic drugs	3.672 (6.795)	16.693 (12.396)	−0.112 (0.613)	7.691
Occupational therapy	0.579 (0.191)	0.794 (0.204)	−0.001 (0.028)	0.717

Notes: This table reports the average effects of CAPS on the number of outpatient procedures performed in the public system per 10,000 people. The first panel reports results for each type of mental health practitioner, and the second shows results for two selected procedures that remain comparable over time in the data. Standard errors in parentheses are computed using a municipality-level clustered bootstrap. Controls include municipality GDP per capita, PBF spending per capita, a series of indicators for age-by-gender population bins, state $\times$ year fixed effects, and linear trends of pretreatment characteristics. Pretreatment municipality characteristics included are: Theil index, poverty rate, unemployment rate, illiteracy rate, share of rural population, log of population, log of social spending, number of mental health providers, number of mental health offices, municipality area, altitude, distance to capital, temperature, and rainfall. Columns 1 and 2 report the average effect after two years and after five years, respectively. Column 3 reports the placebo effect. Column 4 shows the mean of each variable at baseline—i.e., one period before the event.

hospitalizations (> 30 days). Based on average effects, long-stay hospitalizations account for 65.5 percent of the effect. However, the impact takes more time to reach stability. It goes up to 80 percent if we consider only the effect after five years. Turning to specific causes, we see that our results are driven by the reduction of admissions among people with schizophrenia (86.6 percent), which is the only diagnosis for which we find significant effects.

Our findings indicate that CAPS indeed drive a change in the equilibrium of mental health care delivery from inpatient to outpatient care. However, combined with our previous evidence on the profile of people treated at CAPS, our results indicate that most patients diverted from inpatient care are those with schizophrenia no longer staying long-term in hospitals. As already discussed, CAPS are usually ill-equipped to deal with this type of patient.

C. Public Spending

Next, we turn to the impact of CAPS on public spending (Table 4 and Figure 6). In line with our previous results, we find strong and persistent effects on federal

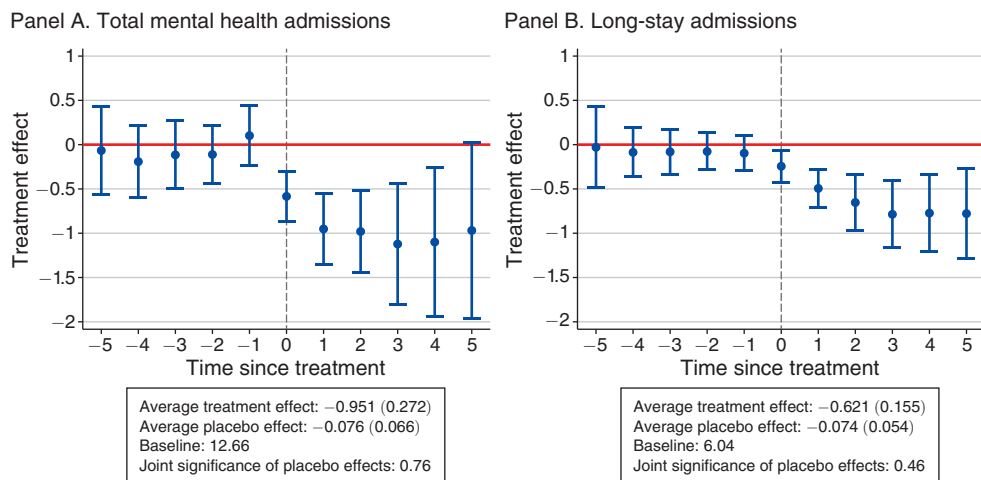


FIGURE 5. EFFECTS OF CAPS ON MENTAL HEALTH HOSPITALIZATION: ADMISSIONS PER 10,000 PEOPLE

(continued)

expenditures with mental health hospital admissions: an average reduction of 15.7 (s.e., 0.04) percentage points. We see no reduction in federal spending on other hospitalizations, suggesting that the reform does not save money elsewhere in the inpatient health care system.¹⁸ Moreover, we find no detectable effects on local health expenditures. These results align with our expectations: most money spent on psychiatric admissions comes from the federal government; municipal governments receive subsidies along with CAPS. Therefore, CAPS does not push local governments to spend more.

With these estimates and the data on total expenses with CAPS, we can assess whether the net financial result is positive or negative. Based on aggregate information provided by the government (Ministério da Saúde do Brasil 2015), the policy's total cost from 2002 to 2014 is 950 million BRL. During the same period, we have 11,545 municipality-years with CAPS. Using the average effect estimated for federal spending in mental health hospitalizations, we have a total cost reduction of 720 million BRL. Hence, the net result is a cost of 230 million BRL. Therefore, from a purely financial point of view and in our particular context, more money is spent with CAPS, contrary to what proponents of the psychiatric reform still argue.

Our results align with descriptive evidence that federal mental health spending increased significantly after the Brazilian psychiatric reform, despite a notable shift in composition favoring community-based instead of inpatient care (Desinstitute and Weber 2021). However, it is important to stress that the financial aspect is only part of the motivation for reform. Data limitations preclude an evaluation of the total

¹⁸ In unreported results, we also see no effects on diabetes and heart attacks, diseases with higher prevalence on the mentally ill population (Ridley et al. 2020), nor on hospitalizations for other causes. These results are in line with our null result on total spending on overall hospitalizations.

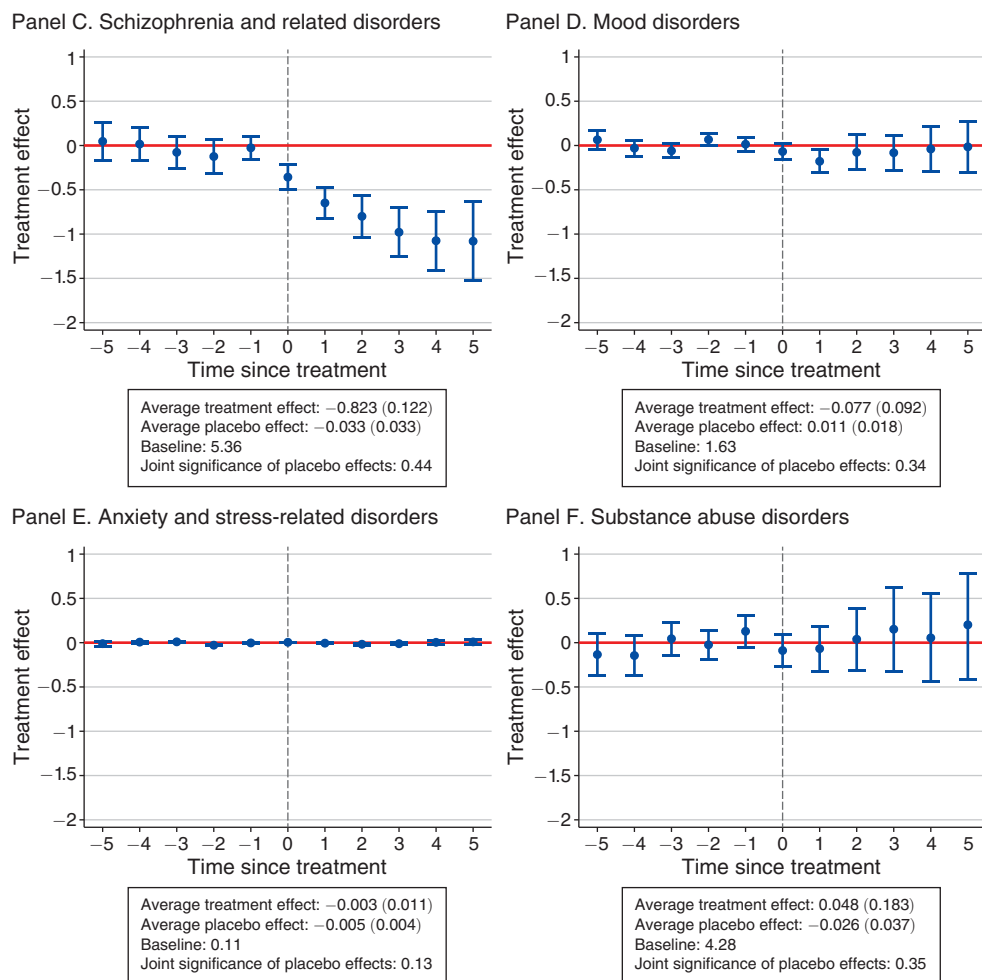


FIGURE 5. EFFECTS OF CAPS ON MENTAL HEALTH HOSPITALIZATION: ADMISSIONS PER 10,000 PEOPLE (*continued*)

Notes: This figure plots 95 percent confidence intervals computed with a municipality-level clustered bootstrap and dynamic and placebo DID estimators for the CAPS' effects on the number of hospital admissions per 10,000 people due to mental and behavioral disorders (ICD-10 F00-F99)—panel A, long-stay hospital admissions due to mental and behavioral disorders—panel B, and hospital admissions by groups of cause: schizophrenia and related disorders (F20-F29)—panel C, mood disorders (F30-F39)—panel D, anxiety and stress-related disorders (F40-F48)—panel E, and substance abuse disorders (F10-F19)—panel F. Long-stay hospitalizations are defined as those in which the patient is hospitalized for more than 30 days. Placebo DID estimators estimate the CAPS' effects had the treatment occurred in a placebo, pretreatment period. They use a varying baseline period—the one immediately before the placebo treatment date—so we do not normalize relative to a unique period. Controls include municipality GDP per capita, PBF spending per capita, a series of indicators for age-by-gender population bins, state $\times$ year fixed effects, and linear trends of pretreatment characteristics. Pretreatment municipality characteristics included are: Theil index, poverty rate, unemployment rate, illiteracy rate, share of rural population, log of population, log of social spending, number of mental health providers, number of mental health offices, municipality area, altitude, distance to capital, temperature, and rainfall. The Average Treatment Effect computes a weighted average of the dynamic estimators, giving to each estimator a weight proportional to the number of switchers the estimator applies to. In parentheses, standard errors are computed with a municipality-level clustered bootstrap. Baseline indicates the sample mean values for those treated in the first pre-CAPS period.